

Automated Banking Website Using Agile Methodology

1. Abstract

Banking systems play an important role in daily life. Traditional banking requires customers to visit the bank physically for services such as account management, money transfer, and balance enquiry. This process is time-consuming and inconvenient.

An **Automated Banking Website** allows customers to access banking services online. Users can register, login, view account details, transfer money, pay bills, and check transaction history from anywhere. All data is stored securely in a centralized database.

This project is developed using **Agile methodology**. Agile divides the project into small development cycles called sprints. Each sprint includes planning, designing, coding, testing, and review. Agile helps in faster development and allows changes based on user requirements.

The proposed system reduces paperwork, saves time, and improves customer satisfaction. It is secure, user-friendly, and reliable. In the future, advanced features like mobile banking apps, biometric login, and AI-based fraud detection can be added.

2. Introduction

2.1 Introduction

An Automated Banking Website provides online banking services to customers through the internet.

2.2 Problem Identification

- Manual banking is time-consuming
- Limited working hours
- Risk of human error

2.3 Need of the Project

- 24×7 banking services
- Faster transactions
- Secure online access

2.4 Project Scheduling

The project is developed in multiple sprints using Agile methodology.

2.5 Objectives

- Provide online banking services
- Manage customer accounts securely
- Improve banking efficiency

3. Software Requirement Specification

3.1 Purpose

To develop an automated banking system for online banking operations.

3.2 Scope

Used by bank customers and administrators.

3.3 Hardware / Software Requirements

Hardware

- Computer / Laptop
- Minimum 4 GB RAM

Software

- Operating System: Windows / Linux
- Programming Language: Java / Python
- Database: MySQL

3.4 Tools

- VS Code / Eclipse
- Web Browser
- Database Server

3.5 Software Process Model

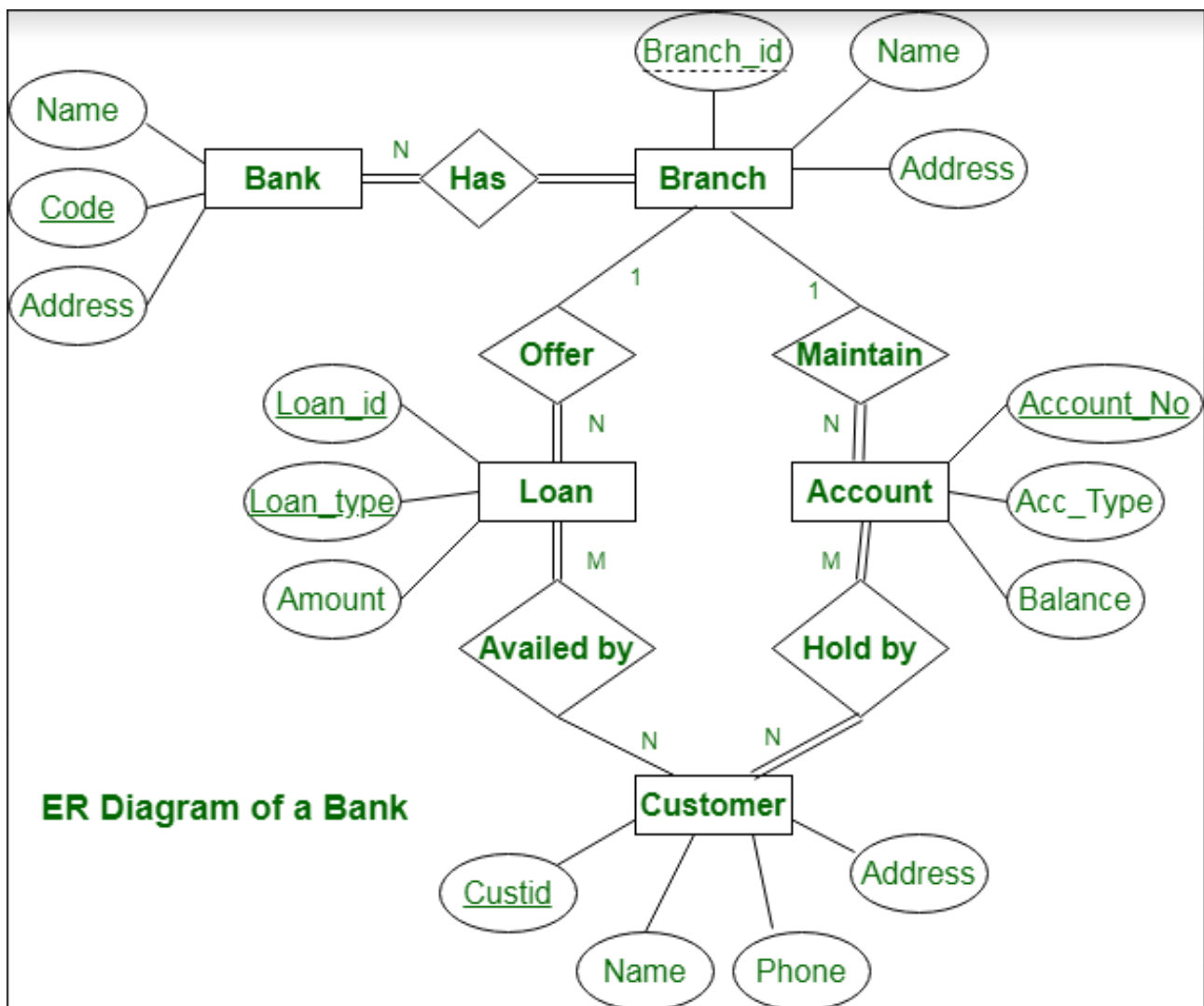
Agile Software Development Model.

4. System Design

4.1 Data Dictionary

Field Name	Description
CustomerID	Unique customer ID
AccountNo	Bank account number
Name	Customer name
Balance	Account balance
TransactionID	Transaction number

4.2 ER Diagram (REAL)



Entities to draw:

- CUSTOMER (CustomerID, Name, AccountNo)
- ACCOUNT (AccountNo, Balance, Type)

- TRANSACTION (TransactionID, Amount, Date)

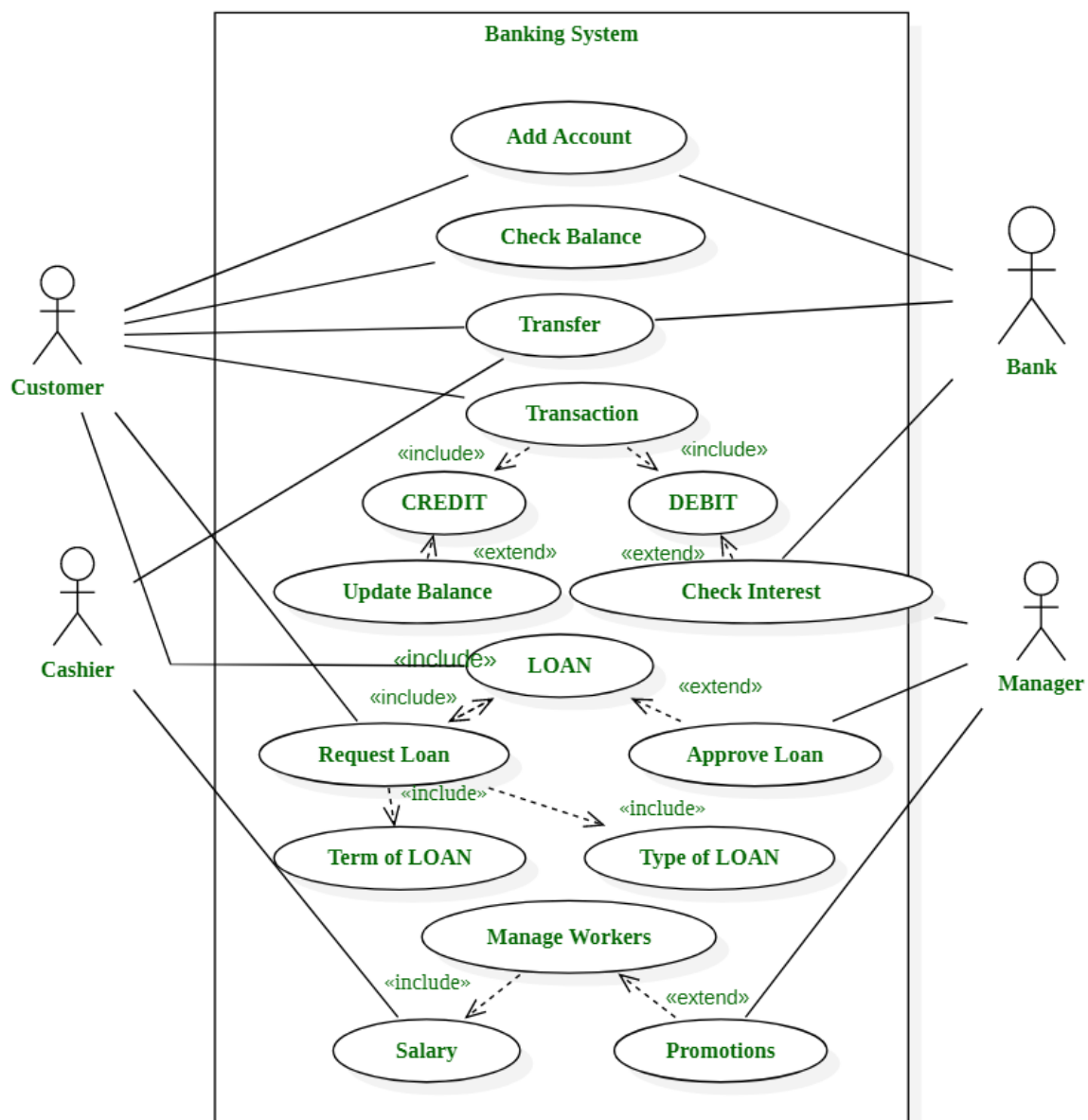
4.3 DFD – Level 0 (REAL)

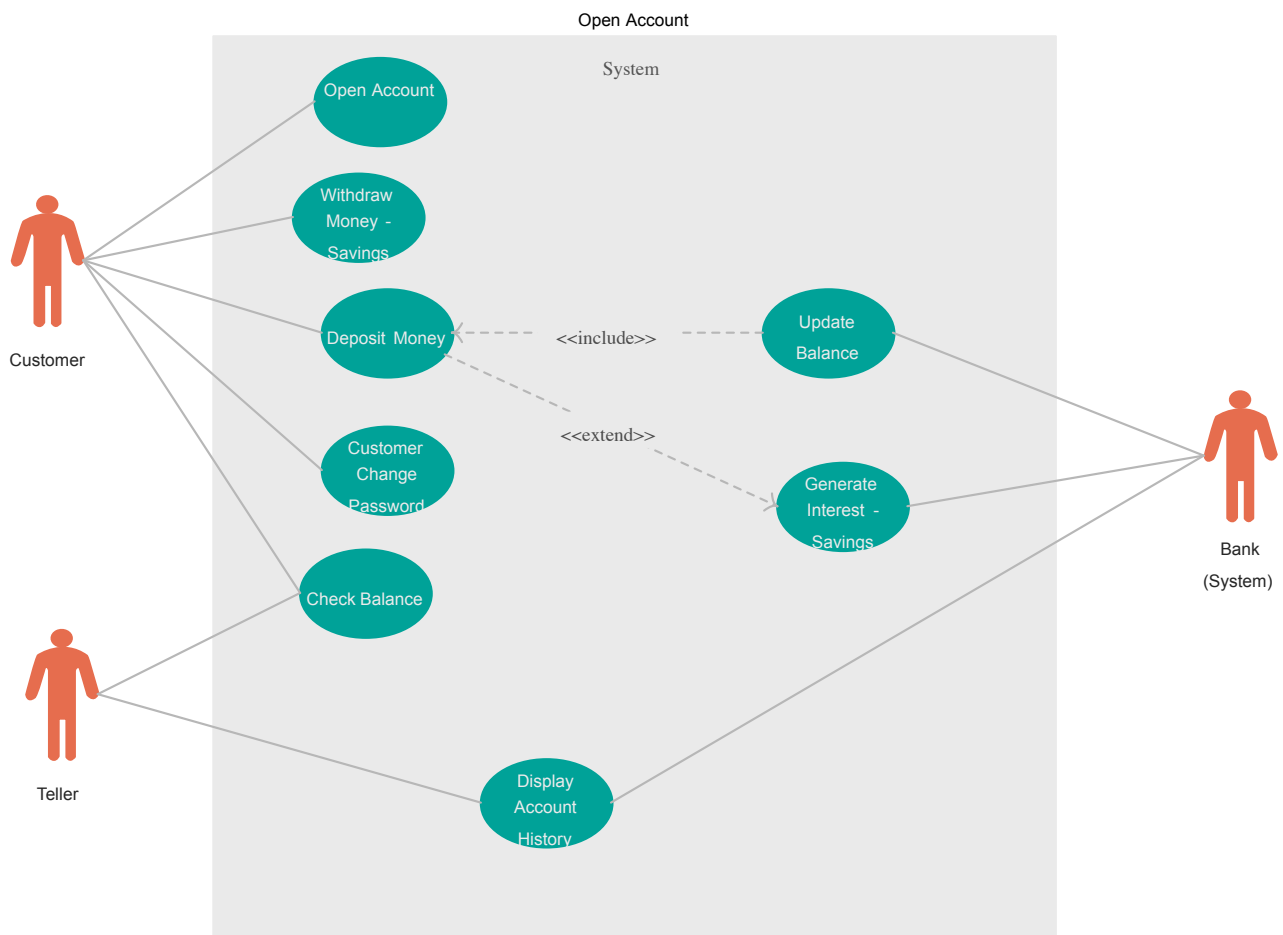
Flow:

Customer → Banking System → Transaction Details

System ↔ Database

4.4 Use Case Diagram (REAL)





Use Cases:

- Login
- View Balance
- Transfer Money
- Pay Bills
- View Transactions

4.5 System Flow Chart (REAL)

Flow:

Start → Login → Select Service → Perform Transaction → Logout → End

5. Implementation

5.1 Program Code

The automated banking system is implemented using Java/Python with MySQL database.

5.2 Output Screens

- Login Page
- Account Dashboard
- Transaction Page

6. Testing

6.1 Test Data

- Valid login credentials
- Invalid transaction data

6.2 Test Result

All banking modules work correctly and produce accurate results.

7. User Manual

7.1 How to Use

1. Login to banking website
2. View account details
3. Perform transactions
4. Logout securely

7.2 Screen Layout

Simple, secure, and easy-to-use interface.

8. Project Applications and Limitations

Applications

- Online banking systems
- Financial institutions
- Digital payment platforms

Limitations

- Internet dependency
- Cyber security risks

9. Conclusion and Future Enhancement

The Automated Banking Website provides fast and secure online banking services. Agile methodology ensures flexibility and efficient development. Future enhancements include mobile banking apps, biometric authentication, and AI-based fraud detection.

10. Bibliography & References

- Software Engineering Textbook
- Agile Methodology Notes
- Online Banking Technology Articles